

Dynamics as a Scale: A Homogeneous Survey of Stability-Derived True-Mass Limits for Non-Transiting Radial-Velocity Exoplanets

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ABSTRACT

Radial-velocity detections constrain only the minimum mass $m \sin i$ of a planet; the true mass can exceed it by an arbitrary factor for face-on orbits. In multi-planet systems, however, long-term dynamical stability bounds that factor: heavier, more face-on configurations perturb their neighbors strongly enough to self-destruct within the system’s lifetime. Discovery papers occasionally apply this constraint to individual systems with heterogeneous methods; no uniform survey exists. We present a homogeneous N -body stability survey of 56 multi-planet RV systems drawn from the NASA Exoplanet Archive, sweeping each system’s line-of-sight inclination and sampling all orbital parameters within their published uncertainties. We validate the pipeline against GJ 876, recovering the literature stability constraint ($i > 25^\circ$ vs. published $i \gtrsim 20^\circ$). We report 95% credible true-mass upper limits under an isotropic orientation prior for 123 planets in 53 systems: 61 planets in 23 systems are tightened below 3.0 times their minimum masses (median limit $2.68\times$ among them), and 25 giant planets are certified below the deuterium-burning limit by dynamics alone. The pipeline additionally documents, and guards against, three classes of catalog-parameter pitfalls that dynamical consistency exposes, including published eccentricity point-estimates incompatible with the survival of their own systems.

Keywords: Exoplanet dynamics — Exoplanet catalogs — N-body simulations — Radial velocity

1. INTRODUCTION

The radial-velocity method measures a planet’s gravitational reflex along the line of sight only, leaving the orbital inclination i unconstrained and delivering the minimum mass $m \sin i$ rather than the mass. For an isotropic distribution of orientations the correction is usually modest — the median of $1/\sin i$ is 1.15, and 95% of orientations imply a true mass below $3.2 m \sin i$ — but for any individual planet the catalog value is a floor, not a measurement. The consequences propagate into everything built on RV masses: mass functions and occurrence rates inherit a one-sided bias, the planet/brown-dwarf boundary is porous for every giant whose $1/\sin i$ could exceed $\sim 13 M_{\text{Jup}}/(m \sin i)$, and models of individual systems’ formation and composition rest on lower limits treated, in practice, as values.

In multi-planet systems, dynamics supplies the missing upper bound. A coplanar system tilted toward face-on hides planets that are heavier by $1/\sin i$; heavier

planets perturb one another more strongly; and sufficiently heavy versions destabilize on timescales far shorter than the systems’ ages. Configurations that cannot survive to the present are excluded, so the survival requirement converts each system’s continued existence into a ceiling on its planets’ true masses. The idea is not new: discovery papers have long applied stability cuts to individual systems (GJ 876 being the canonical example; e.g., [Goździewski 2002](#) for 47 UMa; [Vogt et al. 2014](#) for HD 141399; [Basant et al. 2025](#) for Barnard’s star), and [Tamayo et al. \(2021\)](#) developed stability-constrained characterization into a general method, applied to the transiting system Kepler-23. What does not exist is uniformity. [Laskar & Petit \(2017\)](#) classified 131 systems by AMD-stability but did not invert stability into per-planet mass limits; [Volpi et al. \(2019\)](#) constrained mutual inclinations of ten RV systems with first-order secular theory; [Qin et al. \(2025\)](#) propose breaking the degeneracy through von Zeipel–Lidov–Kozai cycle monitoring in compact binaries; the remaining literature is a scatter of single-system analyses with mutually incompatible integrators, instability criteria, timescales, and statistical definitions. No two

of their mass limits are comparable, and no population statement can be assembled from them.

This paper presents the first homogeneous survey: one pipeline, one instability criterion, one statistical definition, applied to every qualifying multi-planet RV system in the NASA Exoplanet Archive, with orbital parameters sampled within their published uncertainties and the machinery validated against systems whose stability constraints are independently known. We report 95% credible true-mass limits under an isotropic orientation prior for the planets of 56 systems, a byproduct catalog of parameter-consistency flags, and — as a methodological contribution in its own right — three classes of catalog-parameter pitfalls that dynamical auditing exposes (§3.4).

2. SAMPLE SELECTION

We select systems from the NASA Exoplanet Archive’s composite parameters table (`pscomppars`, accessed 2026 July 9–13) requiring at least two planets, periods and masses for all of them, a stellar mass, and at least one non-transiting planet whose mass provenance is $m \sin i$ — the planets the survey can constrain. Because dynamical interactions weaken steeply with separation, we further require at least one adjacent pair with a period ratio below 6. This yields 148 qualifying systems.

Each system is ranked by an analytic pre-estimate of where its ceiling should lie: the swept mass factor f_{crit} at which the most fragile adjacent pair, with masses scaled by f and separations measured in mutual Hill radii (eccentricity-reduced), reaches the two-planet overlap criterion $\Delta = 2\sqrt{3}$ (Gladman 1993). Multi-planet systems destabilize at wider spacings, so f_{crit} is conservative. The survey band takes $f_{\text{crit}} \in (1.05, 60)$ with $e_{\text{max}} \leq 0.75$ and $P_{\text{inner}} \geq 1.5$ d — 56 systems. The excluded $f_{\text{crit}} \approx 1$ block is dominated by resonance-protected pairs whose survival depends on phase coherence that composite catalog parameters cannot supply (§3.4, §6); it is deferred rather than constrained. Table 1 gives the full disposition of all 148 systems: 53 surveyed, 3 flagged, and 92 excluded (42 resonance-protected, 48 too weakly interacting for dynamics to bite at any allowed mass, and 9 with sub-1.5-day inner periods, categories overlapping).

3. METHODS

3.1. Configuration Sampling

A coplanar system tilted rigidly with respect to the line of sight leaves its internal dynamics unchanged while every $m \sin i$ mass scales as $1/\sin i$; we therefore sweep i over a 17-point grid from 90° (edge-on, masses at their

floors) to 5° (masses at $11.5\times$ the floors), holding planets with transiting or astrometric true masses fixed. At each grid point we integrate 20 independent realizations, drawing the stellar mass and every planet’s mass, period, eccentricity, and periastron longitude from truncated normal distributions built on the symmetrized catalog 1σ uncertainties (point values where errors are unreported; eccentricities truncated to $[0, 0.95]$). Unknown angles are drawn uniformly. Orbital phases are randomized for composite-catalog systems and taken from the fit’s time of periastron only for single-reference parameter sets, for the reasons documented in §3.4.

3.2. Integration and Instability Criterion

Realizations are integrated with REBOUND’s WHFast (Rein & Liu 2012; Rein & Tamayo 2015) with timestep $dt = \min_p P_p(1 - e_p)^{3/2}/20$, resolving the fastest perihelion passage. Instability is a close encounter within the sum of Hill radii (particles carry their Hill radius as collision radius) or escape beyond 20 times the widest semi-major axis. Integration time is uniform in inner-planet orbits — 10^5 and 10^6 for the two survey depths — making the effort per system architecture-independent; convergence between the depths is examined in §4.3. Sensitivity of the final limits to these choices is quantified on a six-system representative subset spanning the survey’s architectures: halving dt changes 95% credible limits by ≤ 0.03 in the mass factor; collision radii scaled to $0.5\times$ and $0.25\times$ the Hill radius change them by ≤ 0.08 ; replacing exact coplanarity with mutual inclinations drawn from Rayleigh distributions of $\sigma = 2^\circ$ and 5° changes them by ≤ 0.4 . None approaches the survey’s grid resolution in i .

3.3. Statistical Definitions

Each system yields a survival curve $S(i)$, the fraction of realizations surviving to t_{max} at inclination i . Because parameter draws can be fragile at any tilt, constraints are defined relative to the edge-on baseline: systems with $S(90^\circ) < 0.5$ are reported as parameter-consistency flags — statements about the published parameters, not mass limits. For unflagged systems the primary statistic is the 95% credible limit on the mass factor under an isotropic orientation prior (uniform in $\cos i$), with posterior density proportional to $S(i)$: the factor below which 95% of the surviving posterior lies. With no dynamical information this limit is 3.20 for every RV planet; the survey’s information content is the tightening below that value. A secondary threshold ceiling (smallest factor with $S < 0.5 \times S(90^\circ)$) is tabulated for comparison with prior single-system work.

3.4. Catalog-Parameter Pitfalls Guarded Against

Three recurring pitfalls in catalog parameters, each encountered because it broke an analysis during survey development, are guarded against as follows.

1. Eccentricity point-estimates dynamically inconsistent with system survival (Barnard’s star: $e = 0.03\text{--}0.08$ kills the system in kyr; e drawn within errors resolves it — independently established by [Basant et al. 2025](#), whose fits favor $e < 0.02$). Guard: sample within uncertainties, never integrate point values alone (§3.1).
2. Composite tables (`pscomppars`) mix parameters across incompatible fits — fatal for phase-coherent (resonant) initialization. Guard: randomize phases for composite parameters; use fit phases only from single-reference sets (§3.1).
3. Fit epochs stored in the time-of-periastron column (GJ 876; [Rivera et al. 2010](#)): read naively, the column implies all planets at periastron simultaneously. Guard: reject identical periastron times across a system before using any phase information.

4. VALIDATION

The pipeline is validated against two systems whose stability constraints are independently known, chosen to bracket the two dynamical regimes the survey encounters: a resonance-protected system whose survival depends on its orbital phases (GJ 876) and a near-resonant system that is stable regardless of them (HD 45364). A third, external cross-check appears in §5.1: for Barnard’s star, the survey’s ceilings agree with the independent stability-derived masses of [Byrne et al. \(2026\)](#).

4.1. GJ 876: The Phase-Protected Anchor

GJ 876 is the canonical stability-constrained system. The Archive’s default solution is the coplanar dynamical fit of [Rivera et al. \(2010\)](#) at $i = 59^\circ$: true masses, with the phases of the 4:2:1 Laplace resonance at their libration center. We reconstruct the equivalent minimum-mass system by scaling the fitted masses by $\sin 59^\circ$, fix the mean anomalies at the fit epoch to the published values (their Table 3 — the Archive’s time-of-periastron column stores the epoch itself for this reference, artifact Class 3 of §3.4, so phases cannot be derived from it), and sweep a 13-point inclination grid with phases held fixed (5 draws per point, 5,000 yr per realization $\approx 10^6$ inner-planet orbits).

The validation makes three independent demands (Figure 1). First, the published configuration, entering the sweep at mass factor $1/\sin 59^\circ = 1.17$, must survive:

it does. Second, the recovered constraint must match the literature: survival is zero at every grid inclination at or below 25° and recovers above it, giving $i > 25^\circ$ ($m < 2.37 \times m \sin i$) against the published $i \gtrsim 20^\circ$ — consistent, and conservative in the safe direction. Third, the same masses with randomized phases must not reliably survive, or phase protection would be irrelevant to the problem: survival collapses to 12%. The resonance protection is real, and the phase-constrained initialization captures it.

4.2. HD 45364: The Phase-Agnostic Anchor

HD 45364 anchors the opposite regime. Its two giants sit in a 3:2 commensurability with low eccentricities ([Correia et al. 2009](#)), and the [Li et al. \(2022\)](#) solution is stable without phase assistance: initialized with the fit’s own phases, the published configuration survives the full 10^5 yr integration, and randomizing the phases leaves survival at 100% — the constraint owes nothing to resonance protection. The survival cliff is correspondingly smooth rather than resonance-sharpened: complete survival for $i \geq 45^\circ$, declining to zero by $i \approx 8.5^\circ$ (17-point grid, 10 draws per point), integrating under the isotropic prior to a 95% credible limit of $m < 2.29 \times m \sin i$. Together the anchors bracket the survey population — one ceiling that exists because of its phases, one indifferent to them — and the same pipeline recovers both with no per-system tuning.

4.3. Depth Convergence

A survey whose limits depended on integration length would be reporting its clock, not the dynamics. Every system therefore runs at two depths, 10^5 and 10^6 inner orbits, with identical parameter draws (seeds keyed to the system name), and the threshold ceilings are compared. Deeper integration can only reveal instability, never undo it, so ceilings must move one way — and they do: among the 53 unflagged systems (flagged systems excluded at either depth), 9 ceilings tightened, 5 held exactly, none loosened, and 7 systems acquired a ceiling only at 10^6 (Figure 3). This is the third consecutive survey generation with strictly monotone behavior, and it is what makes the reported limits conservative: a longer clock can only strengthen them (§6). **[PENDING: 10^7 -orbit extension of the 16 non-converged systems — run launched 2026-07-14.]**

5. RESULTS

5.1. The Catalog

Of the 56 surveyed systems, 53 pass the edge-on consistency test and yield 95% credible mass limits for 123 non-transiting planets (Table 2; Figure 2). Sixty-one

planets in 23 systems are dynamically tightened below $3.0\times$ their minimum masses — meaningfully inside the $3.20\times$ that isotropy alone guarantees — with a median limit of $2.68\times$ among them. Twenty-five giants whose face-on orientations would have exceeded the deuterium-burning limit are certified planetary at 95% credibility. Individual highlights: Barnard’s star’s four sub-Earths are capped at $2.18\times$ their minimum masses ($0.42\text{--}0.73 M_{\oplus}$) — consistent with, and independently confirming, the stability-derived masses of $0.19\text{--}0.84 M_{\oplus}$ in [Byrne et al. \(2026\)](#); 47 UMa at $2.35\times$; HD 141399’s four giants at $2.71\times$; and K2-18 c, the non-transiting companion of the habitable-zone sub-Neptune K2-18 b, at $2.87\times \approx 21.6 M_{\oplus}$, excluding a giant-planet identity. (The isotropic prior is conservative for K2-18 c: because K2-18 b transits, only a small mutual inclination is required for c to avoid transiting, so near-edge-on orientations are likely a priori; [Cloutier et al. 2017](#).) The tightest factor in the catalog is a caution against conflating the two headline statistics: TYC 1422-614-1’s pair is capped at $2.01\times$ — the survey’s strongest relative constraint — yet planet c’s minimum mass of $10 M_{\text{Jup}}$ puts even that ceiling at $\approx 20 M_{\text{Jup}}$, above the deuterium-burning limit. A tight mass factor and a certified planetary identity are independent properties.

5.2. Population Statement

The cumulative distribution of credible limits (Figure 4) answers the survey’s title question: exactly half the qualifying planets (61 of 123) have their permitted mass range tightened beyond geometry, the tightest fifth to below $\sim 2.6\times$, and no limit exceeds the isotropic bound beyond the inclination grid’s discretization (the single nominal exception, TOI-1062 c at $3.22\times$ against the analytic 3.20 , is an unconstrained posterior rendered on the 17-point grid) — dynamics only ever removes orientations. Read as demographics: for packed multi-planet RV systems, the “hidden monster” tail of the minimum-mass distribution is largely a geometric fiction; the surviving configurations concentrate near edge-on masses.

5.3. Parameter-Fragility Flags

Three systems fail the edge-on consistency test in more than half their parameter draws: GJ 667 C, HIP 57274, and Kepler-20. These are data-quality results, not mass limits: the published parameter sets, with their stated uncertainties, do not support long-lived configurations at any tilt. For GJ 667 C this echoes the long-standing dispute over its planet multiplicity; for the others it flags eccentricity solutions that their own dynamics reject. A joint dynamical–RV refit is the indicated follow-up in all three cases.

6. DISCUSSION

Three limitations bound the interpretation of the catalog. First, the sweep is coplanar: the system is tilted as a rigid unit, so mutual inclinations enter only through the sensitivity variants (§3.2), which shift the credible limits by at most ~ 0.4 in the mass factor for dispersions up to 5° . Strongly non-coplanar architectures — precisely those the von Zeipel–Lidov–Kozai literature targets — are outside this survey’s assumptions and their limits should be read as conditional on approximate coplanarity. Second, 10^6 inner orbits is short against Gyr ages. Every depth comparison we have run — three survey generations — moved limits monotonically tighter or not at all, so the reported ceilings are conservative: longer clocks can only strengthen them. The converse claim (that a surviving configuration is stable forever) is never made. Third, the inputs are catalog parameters, sampled within symmetrized published uncertainties rather than refit from radial velocities. The parameter-fragility flags quantify where this matters most; for the flagged systems the correct next step is a joint refit, not a mass limit.

The excluded resonance-protected block (adjacent period ratios within a few percent of small-integer commensurabilities, screened out at selection) is addressable with the machinery validated on GJ 876: single-reference parameter sets carrying fit phases, initialized at the observed libration. Extending the survey across that block would add some of the most tightly packed — and therefore most constrainable — systems in the catalog.

The catalog has immediate uses beyond its headline statement. Planets whose dynamical ceiling sits close to their $m \sin i$ floor are the best targets for Gaia astrometric confirmation, since their permitted mass range is narrowest; planets whose ceiling crosses the deuterium-burning limit are certified planetary without further observation, cleaning brown-dwarf contamination out of the giant-planet mass function; and the parameter-fragility flags hand RV teams a prioritized list of systems whose published solutions their own data no longer support.

7. DATA AVAILABILITY

Code and per-run records are available at <https://github.com/Shawn2highh/dynamass>. Catalogs and per-run records will be archived at Zenodo [**TODO: DOI — mint after RNAAS note acceptance**]. This work used the NASA Exoplanet Archive **pscomp** (DOI: [10.26133/NEA13](https://doi.org/10.26133/NEA13)) and **ps** (DOI: [10.26133/NEA12](https://doi.org/10.26133/NEA12)) tables, accessed 2026 July 9–13; we thank the Archive’s

Table 1. System dispositions (excerpt)

System	Disposition	f_{crit}	$S(90^\circ)$	m_{95} factor
Barnard’s star	surveyed	1.60	0.55	2.18
47 UMa	surveyed	1.34	0.55	2.35
HD 141399	surveyed	3.05	0.85	2.71
K2-18	surveyed	17.92	1.00	2.87
GJ 667 C	flagged	5.71	0.15	...
HIP 57274	flagged	1.23	0.45	...
Kepler-20	flagged	4.25	0.40	...
24 Sex	excluded ^a	1.00	...	...
TOI-500	excluded ^b	32.86	...	...

^aResonance-protected / at the stability limit ($f_{\text{crit}} \approx 1$).

^bInner period below 1.5 d.

NOTE—Disposition of all 148 qualifying systems: 53 surveyed, 3 flagged (edge-on survival < 0.5), 92 excluded with reasons (resonance-protected, weakly interacting, or sub-1.5-day inner period). The full machine-readable table is available as `table1_dispositions.csv`, regenerated by `scripts/make_table1.py`.

maintainers, whose stewardship makes surveys like this possible.

Artificial-intelligence assistance (Claude, Anthropic) was used for code development and manuscript drafting; all results were independently verified and are the author’s responsibility.

Software: REBOUND (Rein & Liu 2012), WHFast (Rein & Tamayo 2015), NUMPY, PANDAS, MATPLOTLIB

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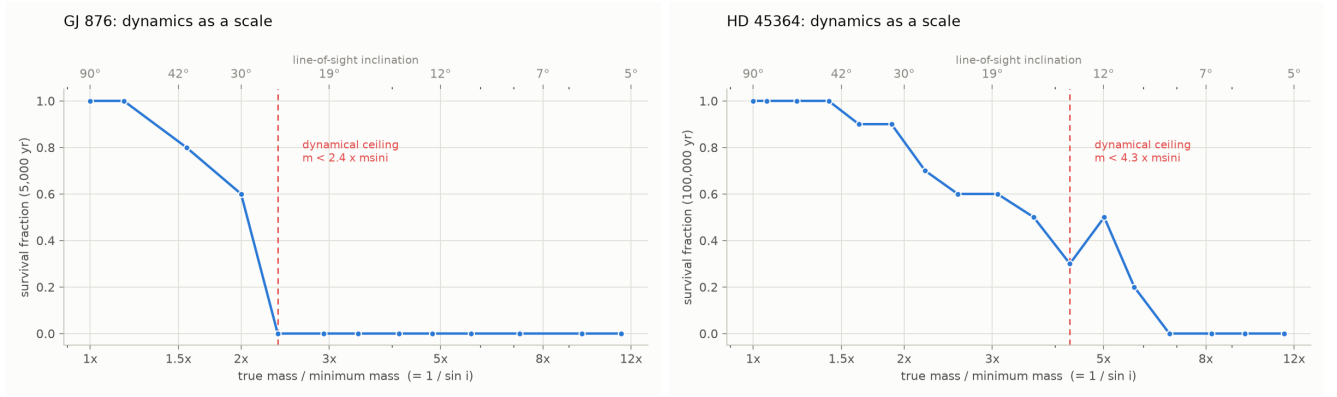
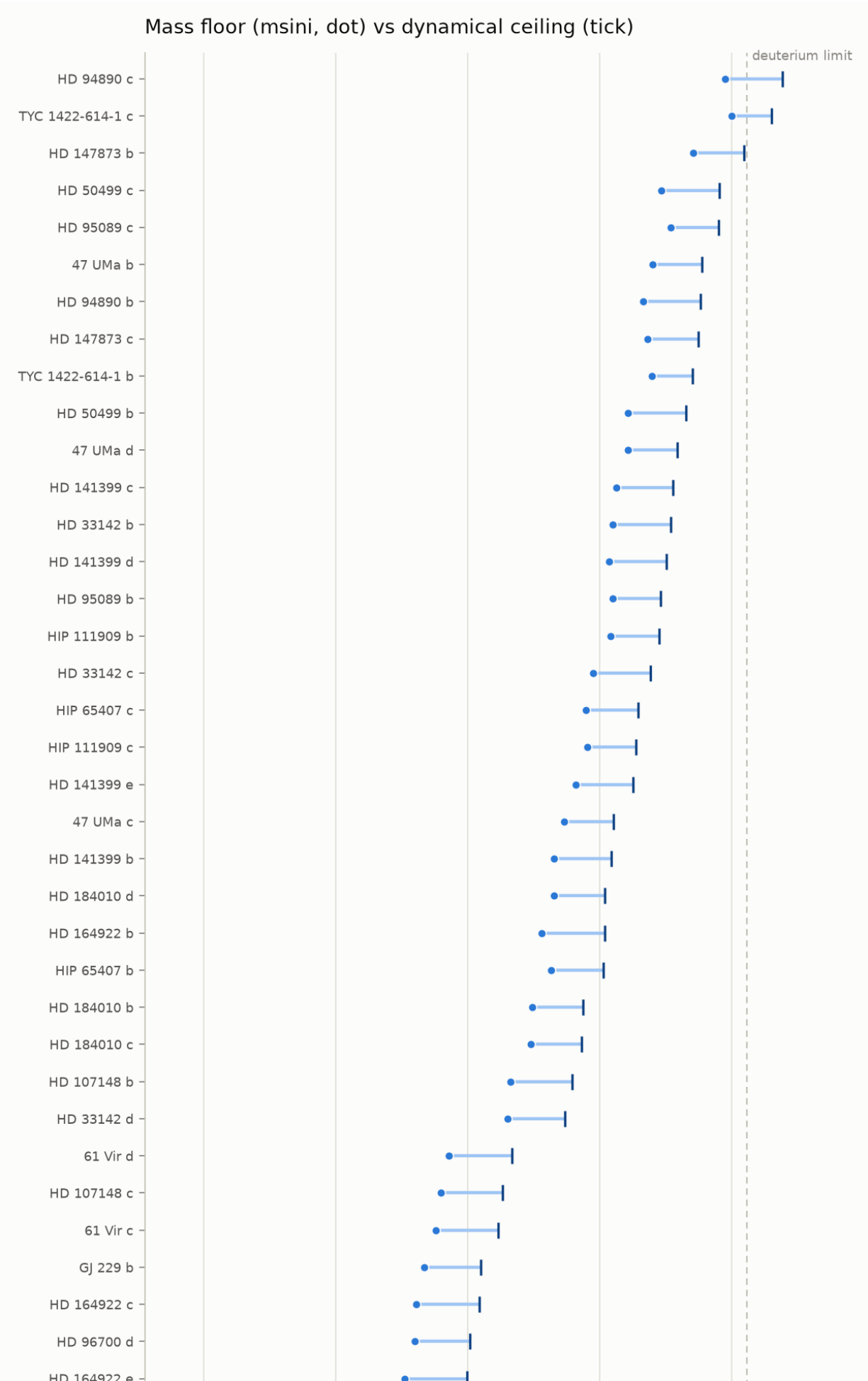


Figure 1. Validation anchors. Left: GJ 876 survival vs. inclination with the Rivera et al. (2010) fit phases held fixed; survival is zero at $i \leq 25^\circ$, recovering the literature constraint. Right: HD 45364 (Li et al. 2022): a smooth, phase-agnostic survival cliff (complete survival at $i \geq 45^\circ$, zero by $i \approx 8.5^\circ$).

Table 2. Per-planet 95% credible true-mass limits (ten tightest)

System	Planet	$m \sin i$ ($M_\oplus$)	m_{95} factor	m_{95} ($M_\oplus$)
TYC 1422-614-1	b	794.5	2.01	1600
TYC 1422-614-1	c	3178	2.01	6400
Barnard's star	b	0.299	2.18	0.65
Barnard's star	c	0.335	2.18	0.73
Barnard's star	d	0.263	2.18	0.57
Barnard's star	e	0.193	2.18	0.42
HD 95089	b	400.4	2.31	923
HD 95089	c	1096	2.31	2527
HIP 111909	b	384.5	2.33	897
HIP 111909	c	257.4	2.33	601

NOTE—The ten tightest of the 123 constrained planets, sorted by 95% credible mass factor. The full machine-readable table (`table2_planet_limits.csv`, regenerated by `scripts/make_table2.py`) additionally lists masses in M_{Jup} , the secondary threshold ceiling, and the inclination floor.



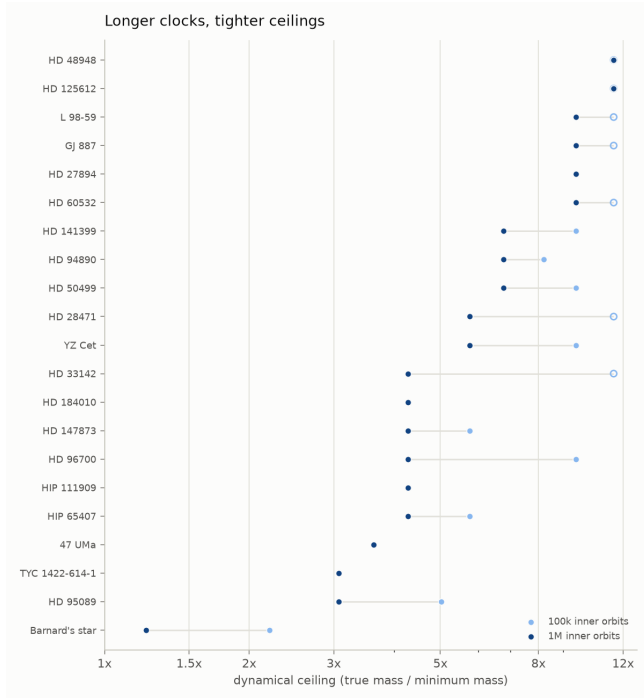


Figure 3. Depth convergence: per-system threshold ceilings at 10^5 vs. 10^6 inner-planet orbits with identical parameter draws. Ceilings only tighten (9) or hold (5); 7 systems acquire a ceiling only at depth; none loosen.

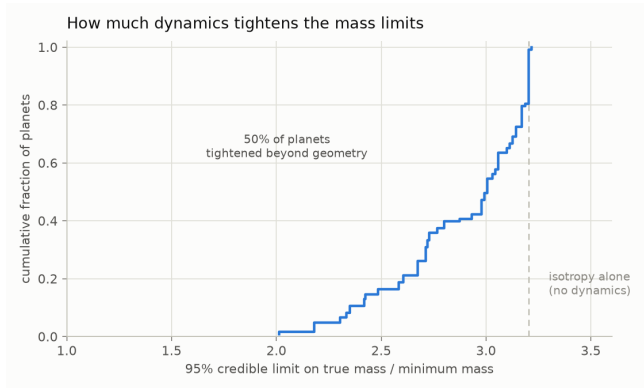


Figure 4. Cumulative distribution of the 123 95% credible mass-factor limits. The dashed line marks the isotropic-prior bound (3.20) that geometry alone guarantees; half the sample is tightened inside it.